

Blair Assessment 3.docx

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Introduction

In this study, information that has been obtained in three consecutive days will be used to assess the likelihood of Alicia's glucose tolerance given her carbohydrate and added sugars intake. The attention is paid to evaluating how fluctuations in macros comprising of carbohydrates, proteins, and fats affect the level of blood glucose in Alicia. Combined with her food diary and glucose tolerance tests, the study will aim to assess the effect of various foods on Alicia's glycemic control. It will assess her macronutrient consumption, portions she is taking, and their relationship to her glucometer readings. The results will enable analysts to understand more about her carbohydrate metabolism and, in general, her metabolism. Furthermore, they will also contain suggestion on the modification of food intake to improve the utilization of nutrients and energy.

Dietary Intake Analysis**Daily Macronutrient Intake**

The dietary log was then reviewed and evaluated for patterns and amounts of carbohydrates, proteins, fats and if necessary, alcohol consumed per day. This process included examination of food and nutrient intake that was acquired from surveys and calculation of MAC and MIC for different diet styles. This division of nutrients by day reflects the detailed levels of nutrients consumed in a day, which is important in assessing an individual's increased nutrient needs or overall nutritional statuses.

This approach of segmenting nutrients allows to see the rationale of the quantities of nutrients that are consumed in a given day. This level of detail is conducive with specific advice-giving when it comes to the respective dietitian client need and general appraisal of overall dietary sufficiency. The actual formulas and detail of how these estimates are worked out will be discussed in the subsequent section in this paper.

Thus, the detailed consideration of a dietary log opens the possibility to define the real quality of the nutrients consumed and reveals how extensively an individual's diet corresponds to their nutritional needs. This specific analysis helps to make proper recommendations to highlight potential improvements in the diet regime and reduce certain aspects that have adverse effects on the body. Thus, focusing on both the MAC and MIC, this analysis aids to both see where nutrient intakes are low or high and enable nutritional intervention to be more effective thereby helping to aid in improving health.

This methodology gives a precise idea about the daily consumption of various nutrients and thus helps in enhancing the overall concept of nutrition planning. The following section will explain further which mathematical calculations are used in these calculations and why these concrete, personalized nutrient ratios are served.

Summary of Daily Intakes

The following table also gives a clear perspective of the daily requirement

of carbohydrates proteins and fats. These macronutrients are divided into separate groups and the portions or amount that an individual should take daily are described. This summary assists in knowing when to take these nutrients considering the fact that they are supposed to be taken all day. To avoid extreme cases in nutrient distribution it is important to enhance distribution mechanisms which see nutrient levels maintaining the recommended daily intake to encourage the engineers of good nutritional health.

Day	Carbohydrates (g)	Protein (g)	Fat (g)
1	257	89	93
2	220	46	55
3	130	46	37
Total	607	181	185
Average (per day)	202.33	60.3	61.7

Average Daily Intake in kcal

For proper identification of kcal in a particular food piece, it requires proper determination of carbohydrates, proteins, and fats present in that particular food piece. Protein and carbohydrates provide calories depending on the grams consumed while fats provide calories dependent on the grams consumed; thus, the knowledge of the type of macronutrient and their caloric values is essential in calculating total calories of the food.

Carbohydrates, proteins and fats are the major groups of nutrients that are required in large proportions by the body. All of the macronutrients supply energy,

although at different levels of efficiency. Namely, carbohydrates and proteins contain 4 thousand of kilocalories per gram, and fats contain 9 thousand of kilocalories per gram. This difference in caloric value is basic when determining the total energetic potential of an item of food. First of all, you have to identify the total grams of macronutrient in a food item as it is illustrated in the following formula. For instance, food sample data may show 20 grams of carbohydrates, 10 grams of proteins, and 5 grams of fats; you will then use these values to arrive at a total of calories.

Needless to say, the calculation for carbohydrates could not be easier. Carry out the operation that involves multiplication of the number of grams of carbohydrates with the caloric value per gram. Here, 20 grams of carbohydrates when converted to kilocalories, using the factor 4 kilocalories per gram, is equal to 80 kilocalories. Likewise, proteins, if the prescribed amount is 10 grams, calculate by multiplying 10 grams of protein by 4 kilocalories per gram, equal to 40 kilocalories. Lastly for fats, 45 kilocalories were obtained by multiplying the specific amount of fat by 9 kilocalories per gram.

When you have decided on the macro nutritional values for each of the food item, addition of these values gives the total caloric value for the food item in question. In the given example, you would add the calories from carbohydrates, proteins, and fats: For non-adipose tissue, the food provides 80 kcal of carbohydrates, 40 kcal of protein, and 45 kcal of fats. Adding it all up, one gets a

total of 165 kilocalories of the particular food item.

When doing this, you are able to determine the exact number of calories of any food from the macronutrient content. This method can also be used to determine the nutritional requirement, promote diet planning and putting into consideration the right diets to adopt so as to achieve the nutritional goals. Appreciating how to analyze the caloric value of the different macronutrients helps to control the dietary intake and contribute to the creation of healthier habits.

Average Daily Energy Intake Calculation

Macronutrient	Amount (g)	Caloric Value (kcal/g)	Total Calories (kcal)
Carbohydrates	203	4	812
Protein	60	4	242
Fat	62	9	559

Here, the amounts of macronutrients are also mentioned and the calories per gram of each macronutrient are multiplied to obtain the total calories of the macronutrient.

Total Average Daily Energy Intake: $812+242+559=1613$

Glucose Tolerance Test Analysis

Absorbance Data and Glucose Concentration

The table below summarizes the results of the glucose tolerance test, providing a comprehensive overview of the measurements and findings. This detailed presentation clearly illustrates the level of glucose tolerance achieved.

Time (mins)	Healthy Person (mmol/L)	Alicia (mmol/L)
0	55765.8	56466
30	88984.6	91510.5
60	99998.7	102154.4
90	92723.6	94730.8
120	66795.7	68273.8
150	67041.5	68968.4
180	66334.9	67803.7

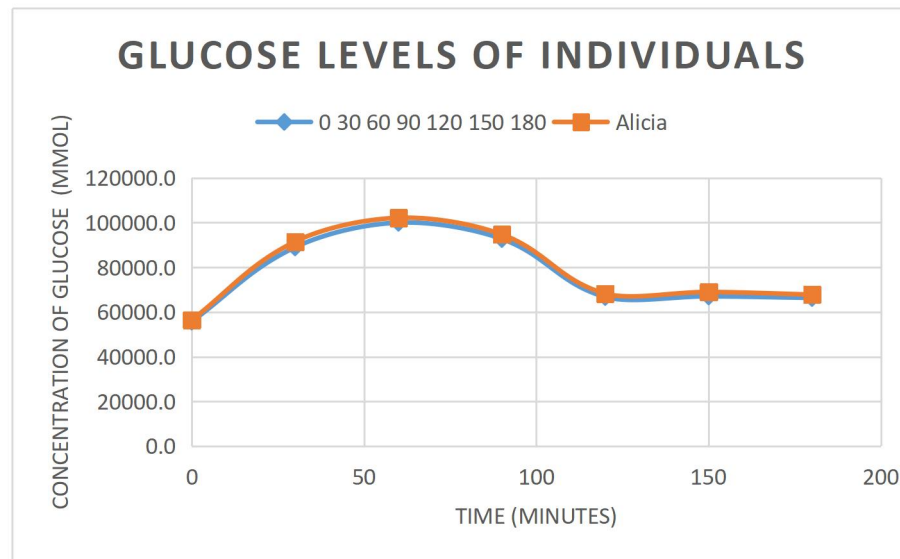
2.2 Graphs

The graph shown above shows the fluctuation of glycogen for normal people and for Alicia. According to the six-hour observational period, ideal blood glucose should remain under 7; however, Alicia's readings are pathologically higher. To start with, the glucose levels increase in Alicia, at its highest level during the 60-minute activity and thereafter it decreases. This pattern may suggest damage in the metabolism of glucose, possibly an early sign of insulin resistance/diabetes.

On the other hand, persons with abnormal glucose metabolism show relatively small but frequent disturbances in glucose concentrations during the day and at night. Alicia's glucose levels however rise for a long time and this

circumstance may indicate that there is poor glucose uptake and transport seen in glucose intolerance and diabetes. This creates a necessity for periodic monitoring for the possibility of other metabolic disorders despite maintaining normal first thing in the morning glucose levels and high post-meal glucose level.

Monitoring tools like GLS and SMBG should be used when treating these conditions and it remains mandatory that people with these conditions get in touch with metabolic physicians to avoid a transfer to diabetes in case the two conditions are not well managed.



Graph 1: Glucose Concentration Over Time (Healthy Person vs. Alicia)

Conclusion

Observing the changes in the frequency of eating and choice of foods in relation with the glucose tolerance test results reveals important information about Alicia's eating behavior and the current status of her metabolism. This analysis relies on her basal metabolic rate and blood glucose levels to assess her meals. A

comprehensive evaluation of the glucose test and the food diary will help determine the various macronutrient components of the client's diet and modify them to improve her metabolism. The detailed analysis presents professional advice that as adapted for her individual needs will improve her metabolic profile.

Dietary Analysis

A brief analysis of Alicia's macronutrient counts over a period of three meals sampled, compared to a checklist of ADA guidelines, indicates that the consumption of macronutrients in Alicia's diet is within the acceptable limit. Thus, Alicia's regular intake per day is 203 calories of carbohydrate, 60 milligrams of calcium, 560 milligrams of iron, 7 grams of protein, and 62 grams of fat. Consequently, her fat intake to be around 2 grams a day is within the dietary guidelines on macronutrient distribution of 45-65% carbohydrates, 10-35% protein, and 20-35% fats. However, without any doubt, Alicia's overall daily caloric intake is still, relatively, high and is approximately 1,615 kcal, which does not stray far from the total energy expenditure upper limit (TEE) for the sedentary AM, who typically whose daily calorie requirement ranges between 1,600- 2,400 kcal. This additional amount of calories, especially from simple sugars and saturated fats, can actually worsen her health condition. Eating starchy and calorie rich products in regular basis which contains processed foods and unhealthy fat leads to high body weight and is most often associated with metabolic diseases. Energy and nutrient rather than calorie partitioning is vital in collaborating healthy weight, and reducing risks associated to chronic diseases.

Glucose Tolerance Test Results

From the results of the glucose tolerance test in Alicia, there has been a rather disturbing trend on her glucose parameters. Her initial glucose level that was assessed was a mere 56mg/dL (or 47mmol/L); a figure that is below normal. But it rose further to 102 mg/dL (60 mmol/L) as the test continued until she completed the test. The glucose values recordings gave the highest value of 15 mmol/L at the 60 th minute and slightly declined to 12 mmol/L after 180 minutes. The consistently high value of glucose right through the test may come with several indications of problems with glucose metabolism. An otherwise normal glycemic changes after eating would show comparatively elevated levels immediately after eating, followed by a drop to normal, low or below; Alicia's Glucose results point at insulin resistance or prediabetic type of glucose metabolism problems.

Recommendations

It was recommended that in order to address the high blood sugar levels of Alicia, she should expand her search for the possible reasons. A considerably more detailed assessment should include new parameters like FBS and HbA1c as these will indicate short-term and, respectively, long-term glycemic control and possible metabolic derangement. Furthermore, embracing of some aspects of life style changes could boost the improvement in glycemic control greatly. This

means the proportional reduction of the portion of high calorie or sugary foods while at the same time increasing the intake of the right portions of good grains, proteins, and good fats. Fitness activities could be done frequently because they affect insulin and glucose. They should consult a registered dietitian or their healthcare provider for advice on individualized meal and physical activities plan. It is also preferred to monitor these aspects repeatedly to detect other health related dangers that might arise from the future complications of macronutrient deficiency.

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